

CÓDIGO:

Question 3. The lattice energy of KBF_4 (10% of the total; 10th, 11th and 12th grades)

3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	Total	Total
4	4	2	8	2	2	2	10	6	40	10%

The lattice energy or net energy (U°) is the energy involved in the formation or rupture of an ionic solid, taking as reference the isolated ions (in the gaseous state). The sign of U° depends on the convention used. From now on, consider that this quantity is negative and refers to the solid formation reaction. The lattice U° values can be estimated from Coulomb's law, assuming that the interaction between the ions that form the crystal is essentially electrostatic. The Born-Mayer equation contains specific empirical-theoretical parameters that depend on both the crystalline structure (spatial arrangement of the particles) of the solid and the electronics of the ions:

$$U^\circ = -\frac{N_A \cdot M \cdot |z_+| \cdot |z_-| \cdot e^2}{4 \cdot \pi \cdot \epsilon_0 \cdot r_0} \left(1 - \frac{\rho}{r_0}\right)$$

The constant ρ has a value of 35 pm for all alkali metal halides; $N_A = 6.0221 \cdot 10^{23}$ is Avogadro's number; M is Madelung's constant and is determined by the arrangement of the ions in the crystal (crystalline structure); z_+ , z_- are the respective charges of the cation and anion; r_0 is the internuclear distance between the two nearest oppositely charged ions; $\epsilon_0 = 8.854 \cdot 10^{-12} \text{ C/(m} \cdot \text{V)}$ is the electrical permittivity in vacuum, and e is the elementary charge of the electron, whose value is $1.602 \cdot 10^{-19} \text{ C}$.

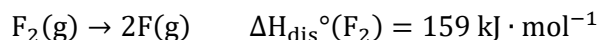
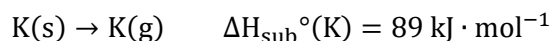
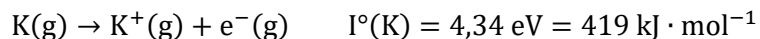
3.1. Use the previous equation to show that the U° of potassium fluoride is close to -800 kJ/mol . Consider that M is 1.7476 and the interion distance in the lattice r_0 is 271 pm. $1 \text{ pm} = 10^{-12} \text{ m}$

3.2. Usually, cations are smaller than anions, and the radii ratio is $r_+/r_- < 1$. This does not occur in the fluorides of the heavier metals. Calculate the value of r_+/r_- in potassium fluoride, knowing that the K^+ ion has an ionic radius (r_+) that is 5 pm greater than the ionic radius (r_-) of F^- .

3.3. There is a slight numerical difference between the lattice enthalpy (ΔH_U°) and U° since the latter is the variation of internal energy during the formation of the ionic solid. Although we will consider that both values are equal throughout this question, complete the following expression that relates both magnitudes for potassium fluoride by applying the First Principle of Thermodynamics.

$$\text{K}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{KF}(\text{s}) \quad \Delta H_U^\circ(\text{KF}) = U^\circ(\text{KF})$$

Thermochemical cycles allow us to find energy values that are difficult to measure. For example, the Born-Haber cycle relates U° to the enthalpy of formation of the ionic solid (ΔH_f°) and different atomic and molecular properties. Consider the following data:



We will use as a convention that the electronic affinity (A° , positive) is the opposite of the enthalpy of electronic gain ($\Delta H_{\text{ge}}^\circ$, negative).

3.4. Use the Born-Haber cycle and show that A° for the fluorine atom is close to 3.5 eV.

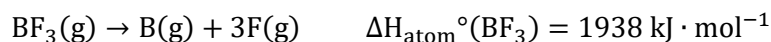
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According to Mulliken, the electronegativity (χ) equals the half-sum of the electron affinity and the ionization potential (in eV). For his part, Pearson established that hardness (η) is equal to the half-difference of the same properties to develop a mathematical basis for his Theory of Soft Hard Acids and Bases.

3.5. Calculate χ and η for the fluorine atom. $\chi(\text{F}) = \frac{I+A}{2}$ $\eta(\text{F}) = \frac{I-A}{2}$

3.6. Using your knowledge of chemical periodicity and the previous expressions, justify why fluorine is a harder species than chlorine.

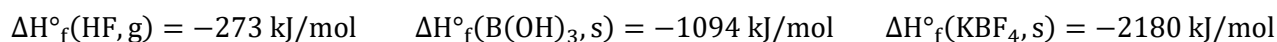
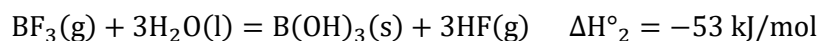
The enthalpy of atomization ($\Delta H_{\text{atom}}^\circ$) can be defined as the heat evolved (at constant temperature and pressure) in breaking all the bonds of a molecule in the gaseous state to obtain the individual atoms that make up the molecule.



3.7. Assuming that the B-F bond is purely covalent, estimate the enthalpy of formation of the gaseous tetrafluoroborate ion ΔH_1° from the above data only.



3.8. Calculate ΔH_3° for the formation of salt KBF_4 using Hess's Law and the following data.



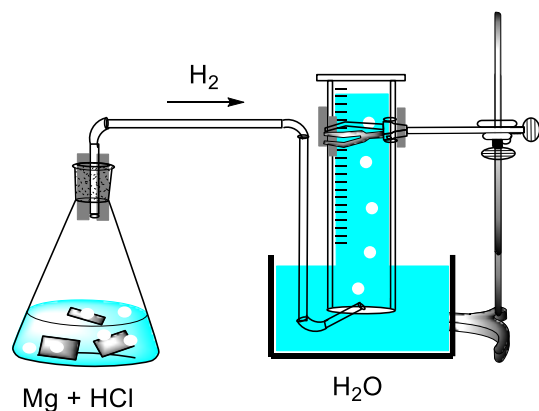
3.9. Calculate U° for KBF_4 salt employing Hess' Law and the following data. If you could not answer items 3.1, 3.7 and 3.8, you may use for your calculations the values $U^\circ(\text{KF}) = -800 \text{ kJ/mol}$, $\Delta H_1^\circ = -650 \text{ kJ/mol}$ and $\Delta H_3^\circ = -450 \text{ kJ/mol}$.



Question 4. Is it chemistry, mathematics or physics? (15% of total; 10th, 11th and 12th grades)

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10	Total	Total
2	2	8	4	10	4	10	4	4	12	60	15%

In the chemistry laboratory of the National Training Center, a kinetic study of the reaction between magnesium and hydrochloric acid was carried out. For this purpose, the equipment shown in the figure was attached. To reduce errors, excess concentrated acid and fragments of different length l of the same tape (width $a = 3.5 \text{ mm}$; thickness $h = 0.15 \text{ mm}$) of metallic magnesium ($A_r = 24.3$; volumetric density $\rho = 1.74 \text{ g/cm}^3$ were used.). Next, the volume of dihydrogen formed as a function of time was determined for different experiments in which the variables temperature, acid concentration and exposed magnesium area were carefully controlled. The data obtained made it possible to calculate the initial reaction rates.



4.1. State the chemical equation of the reaction that occurs.

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4.2. Why do magnesium strips left outdoors turn an ashy color?

4.3. Express the rate of the reaction v_R as a function of the variation of the volume of dihydrogen formed ΔV_{H_2} in a tiny time interval Δt , the temperature T , the atmospheric pressure p_T , the vapor pressure of water p_{H_2O} and the volume of the acid solution V_D (consider this value as constant).

$$v_R = \frac{1}{2} \cdot \frac{dc(HCl)}{dt} \approx \frac{1}{2} \cdot \frac{\Delta c(HCl)}{\Delta t} \quad \text{para } \Delta t \approx 0$$

4.4. Calculate the surface densities σ (g/cm²) and linear densities λ (g/cm) of the magnesium tape used.

4.5. In some experiments, a piece of tape was used so that the total exposed magnesium surface was 70 mm². Calculate: i) the length l of the fragment, ii) the mass of magnesium it contains and iii) the volume (at 25 °C and 1.00 atm) of dihydrogen measured at the end of the reaction and the fragment being consumed.

4.6. Name all the laboratory supplies represented in the figure.

The results obtained in the experiments are summarized below:

Reaction	[H ₃ O ⁺], mol/L	Surface area of Mg, mm ²	Temperature, °C	Initial rate, mol/(L·s)
1	1,00	70	25,0	$1,56 \cdot 10^{-4}$
2	0,200	70	25,0	$5,64 \cdot 10^{-6}$
3	0,500	70	25,0	$3,36 \cdot 10^{-5}$
4	1,00	140	25,0	$2,79 \cdot 10^{-4}$
5	1,00	35	25,0	$7,30 \cdot 10^{-5}$
6	1,00	70	35,0	$2,67 \cdot 10^{-4}$
7	1,00	70	15,0	$1,19 \cdot 10^{-4}$

4.7. Determine the rate law and the constant k_R at 25°C. Round partial reaction orders to the nearest integer value. Say the units of k_R .

$$v_R = k_R \cdot A_{Mg}^{???} \cdot [H_3O^+]^{???}$$

4.8. Assuming that the activation energy is 30 kJ/mol, how many times will the rate of the reaction increase when the temperature is increased from 25 to 35 °C, keeping other variables constant?

4.9. Calculate the density (g/L) of dihydrogen at 25 °C and 1.00 atm pressure.

In their movement, the bubbles accelerate until they reach a maximum speed value when the forces are compensated. Opposing forces act on the bubbles. The thrust (E) favors the upward movement, and its module is numerically equal to the weight (F_g) of the volume of liquid displaced by the displaced bubble. Instead, the weight of the bubble and the friction force (F_d) oppose the movement. The friction force can be calculated using Stokes' Law, knowing that the dynamic viscosity coefficient of water is $\mu = 1.002 \cdot 10^{-3}$ Pa·s.

Fuerza de gravedad $F_g = m \cdot g$

Ley de Stokes $F_d = 6 \cdot \pi \cdot r \cdot \mu \cdot v$

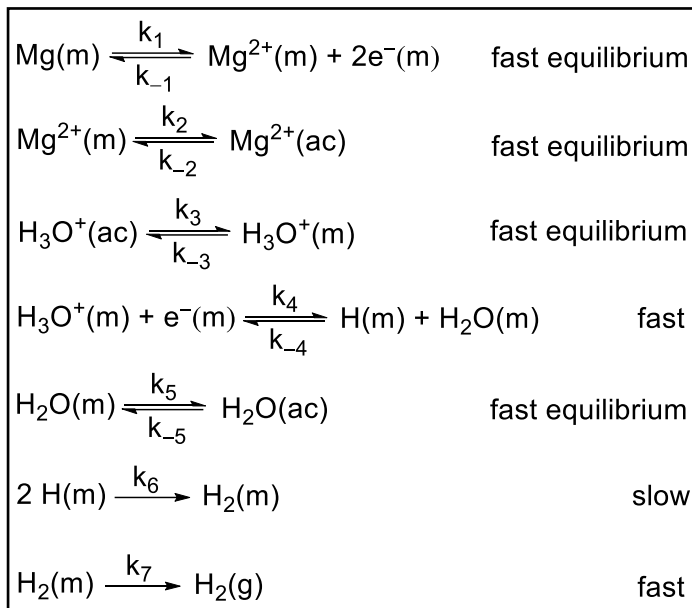
4.10. **Find an expression** and calculate the value of the maximum velocity v_{max} (in m/s) with which dihydrogen bubbles of radius $r = 5.0$ mm move inside the container. $\rho_{H_2O} = 1.00$ g/cm³, $g = 9.81$ m/s²

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Question 4A. (5% of the total; only 12th grade)

4A.1	Total	Total
20	20	5%

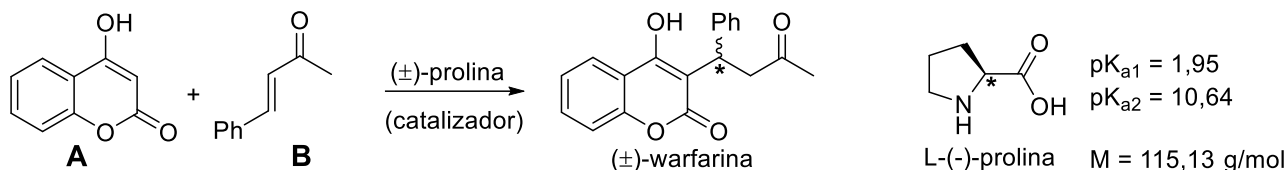
4A.1. For the previous reaction, the following mechanism has been proposed. The aggregation state of the species adsorbed on the metal surface is denoted as “(m)”. It is assumed that Mg^{2+} cations and free electrons exist on the surface of the metal. Consider that the ratio $[\text{Mg(m)}]/[\text{Mg}^{2+}(\text{ac})]$ is proportional to the area of reacted A_{Mg} metal. Use the pre-equilibrium approximation to express the approximate rate law in terms of the individual step constants. Treat the concentration of a species on the metal surface or in solution in an analogous manner.

**Question 5 Warfarin synthesis** (20% of the total; only 12th grade)

5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	5.11	5.12	5.13	5.14	5.15	5.16	Total	Total
6	2	2	2	6	8	2	6	8	6	2	4	4	2	16	4	80	20%

In the Joint Chemical and Biomolecular Synthesis Laboratory “José Luis Mola Garate” (SINTLAB) of the University of Havana (UH), the optimization of the synthesis process of active ingredients of several essential medicines is carried out to implement its production in Cuba and avoid costly imports. One of the substances under study is warfarin, a clotting inhibitor discovered by Karl Paul Link and his team in 1948 at the University of Wisconsin. Warfarin is derived from the natural product dicumarol, which is found in putrid sweet clover and causes death from excessive bleeding of animals that eat the plant. The exciting story of its discovery can be found in the BBC article “How the anticoagulant warfarin, the rat poison that doctors prescribe, was discovered.”

In the laboratory, the synthesis of warfarin is carried out by the reaction between 4-hydroxycoumarin (**A**) and trans-4-phenyl-3-buten-2-one (**B**), catalyzed by the natural amino acid proline.



5.1. 4-Hydroxycoumarin can exist in the form of three tautomers. Why does the tautomer that appears in the diagram exist in a greater proportion? Represent the structures of the two remaining tautomers.

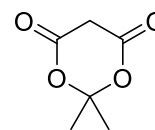
5.2. Name the reaction above and calculate the atomic economy (AE) percent.

5.3. Why is warfarin obtained in racemate form in the above reaction?

5.4. Write the adjusted equation for the condensation reaction that allows you to obtain reactant **B**.

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5.5. 4-Hydroxycoumarin can be obtained in two reaction steps by reacting phenol with Meldrum acid at high temperature, forming acetone and compound **C**, which undergoes an intramolecular Friedel-Crafts reaction in an acid medium. Propose a mechanism for both reactions.



Meldrum' acid

5.6. 4-Hydroxycoumarin gives rise to the diazotization reaction by the most nucleophilic carbon atom in its structure. Represent the synthesis of the simplest possible diazonium salt from benzene and the product **D** formed by its diazotization reaction with 4-hydroxycoumarin.

5.7. Proline in nature occurs mainly in the form of the L enantiomer, shown in the previous scheme. Classify the stereogenic center (*) according to its absolute configuration (R/S) and the proline according to the direction of the deviation of polarized light (right-handed/left-handed).

5.8. Proline exists in a neutral ionic form (zwitterion) in solid state and aqueous solution. In strongly acidic or basic aqueous media, proline is found in cationic or anionic form, respectively. Represent the three charged structures of proline described above.

In the main reaction of the synthesis, a relatively large amount (50 mol%) of proline is used, which performs the double function of base and organic catalyst. Proline deprotonates coumarin **A**, forming a very nucleophilic anion **E**. In contrast, as a product of the reaction between proline and the unsaturated ketone **B**, the formation of a very electrophilic intermediate **F** occurs.

5.9. Represent the resonant structures of the intermediates **E** and **F**. Classify them according to the Theory of Hard and Soft Acids and Bases.

5.10. Calculate the proportion in which the three forms of proline (cation, zwitterion, anion) are found in pure water (pH = 7.00) with respect to the total amount of proline dissolved (α_0 , α_1 , α_2 , respectively). Hint: Consider for your analysis that cationic proline is a diprotic acid.

5.11. The isoelectric point (pI) of a molecule is the pH value for which the predominant form is the neutral one, in the case of proline, the zwitterion. Calculate the isoelectric point of proline.

The partition or partition coefficient (K_D) is the ratio between the equilibrium concentrations of a substance X in the two phases of a mixture of two solvents (Y, Z) immiscible at equilibrium. It is often confused with the distribution coefficient (D), which is the ratio between the total concentrations of the substance, which includes all the ionized forms X_i in which it is found.

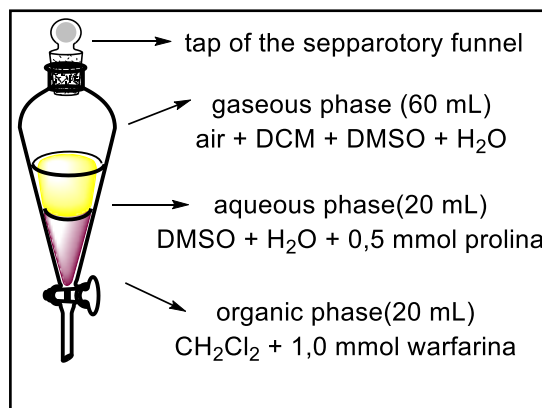
$$K_D = \frac{[X]_Y}{[X]_Z} \quad D = \frac{c(X)_{\text{total en Y}}}{c(X)_{\text{total en Z}}} = \frac{[X1]_Y + [X2]_Y + \dots}{[X1]_Z + [X2]_Z + \dots}$$

5.12. Estimate the partition coefficient of proline between water and dichloromethane if the solubility of proline is 162 g in 100 g of water ($\rho = 1.00$ g/ml) and 1.0 g in 100 g of dichloromethane (DCM, $\rho = 1.33$ g/ml).

5.13. Find an expression for the distribution coefficient of proline in the water/dichloromethane system based only on the constants K_{a1} , K_{a2} , pH and K_D .

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When warfarin synthesis is completed, the crude or mixture product of the reaction is poured onto water, and the resulting suspension is transferred to a separating funnel, where the product is extracted and washed. In one of the experiments at SINTLAB, a researcher used 1.0 mmol of each reactant and 5.0 ml of dimethyl sulfoxide (DMSO; $M = 78.13 \text{ g/mol}$; $\rho = 1.10 \text{ g/ml}$) as a solvent. Subsequently, he added 15.0 mL of water and transferred the resulting suspension to a separatory funnel with a capacity of 100 mL. Under these conditions, all the DMSO dissolves in the water and the product precipitates



due to the change in the polarity of the medium. Then, the researcher added 20.0 ml of dichloromethane ($M = 84.93 \text{ g/mol}$; $\rho = 1.33 \text{ g/ml}$), closed the funnel and shook vigorously to dissolve the product in the organic phase. However, the researcher was not careful, and due to the pressure difference between the gas inside the funnel and the outside air, the funnel tap was thrown off.

Fortunately, he did not lose any of the product and could collect the organic phase. Because the investigator wanted to recover as much warfarin as possible, he performed a second extraction with 20.0 mL of dichloromethane. This time, after shaking the mixture vigorously inside the separatory funnel, he carefully removed the tap and shook again, repeating these steps until the pressures inside and outside the container were equalized. He then let it sit so that the organic and aqueous phases separated completely. Then, he combined the organic phases collected in the previous two extractions (40 ml), washed them with brine, and, after separating them in the funnel, dried the organic phase with anhydrous sodium sulfate. After evaporating all the solvent and leaving the very high-purity solid obtained in a desiccator overnight, he calculated that the final yield of the process was 80%.

5.14. Why is warfarin more soluble in dichloromethane, while proline is more soluble in water?

5.15. Using the data shown in the figure, calculate the pressure of the gas phase inside the separating funnel after reaching equilibrium. Assume that before stirring, the gas phase contained only air (21 mol% dioxygen, 78 mol% dinitrogen, 1 mol% argon). The temperature during the experiments was 25.0°C and the air pressure inside the laboratory was 1.00 atm. The vapor pressure of DCM, water, and DMSO liquids is 46.7 kPa, 3.1664 kPa, and 0.06 kPa, respectively. Consider that Dalton's and Raoult's laws are valid, the volumes of liquids are additive, the ideal behavior of the gas phase and the dissolved solutes are non-electrolytes.

$$\text{Dalton's law} \quad p_{\text{total}} = \sum_i p_i^{\text{gas}} \quad \text{Raoult's law} \quad p_i^{\text{gas}} = x_i^{\text{liquido}} \cdot p_i^{\circ}$$

You must calculate: $p(\text{O}_2)$, $p(\text{N}_2)$, $p(\text{Ar})$, $p(\text{DCM})$, $p(\text{DMSO})$, $p(\text{H}_2\text{O})$ y $p(\text{total})$ in kPa.

5.16. What is the maximum mass (in kg) that the tap of the funnel could have so that the pressure difference between the gas inside the container and the air outside would be able to trigger it? Consider that the tap is shaped like a cylinder with a diameter of 1.0 cm. Neglect the friction force between the tap and the upper part of the funnel. If you did not solve the previous part, assume that the pressure of the gas inside the funnel is $p(\text{total}) = 150 \text{ kPa}$.

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamic	$Q = \Delta U + W$
Zeroth in Celsius's scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs's energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pressure depression	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of the sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of the sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of the circle	$A_{\text{sp}} = \pi \cdot r^2$	Light speed	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of the circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity acceleration on the Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]